Auditing LLM-Based Evidence Screening for Manufacturing Supplier Discovery

Yijiashun Qi
ORCID: 0009-0009-2129-6932

Exploratory Working Paper v0.2

Abstract

LLM-assisted supplier discovery depends on whether retrieved pages support specific business claims, yet a plausible assessment can conceal failures in quotation and adjudication. We audit an evidence-screening workflow on a frozen corpus constructed from 960 U.S. manufacturing queries. Direct acquisition produced 3,383 accessible pages and 6,164 source-capability-state pairs. Separate passes of the same model assessed five producer-support conditions; disputed fields received a third pass. The final workflow assigned 1,148 positive labels (18.62%), 3,799 no labels, and 1,217 unclear labels. Retrospective reconstruction exposes two consequential processing decisions: replacing whole-record adjudication with fieldwise merging restored 321 A/B-agreed no labels, and literal-quote re-audit downgraded 31 of 71 flagged positive rows to unclear. These changes measure output sensitivity and internal consistency, not gains in factual accuracy. We document the observation unit, frozen evidence boundary, and correction sequence, with public code and aggregate artifacts. The contribution is a domain-specific process audit that separates model-label stability, quotation integrity, and external validity. Independent human validation remains necessary before using the labels as a benchmark.

Keywords: LLM evaluation; supplier discovery; evidence screening; adjudication; web-corpus audit; provenance

Study status. A retrospective audit of model outputs on a frozen corpus. Independent human validation and the preregistered entity-resolution audit remain incomplete. This revision is not a validated benchmark or a publication-ready study.

1. Introduction

A manufacturing search may retrieve a company name, a capability description, and a state name without establishing that the company produces that capability in that state. A distributor can describe a product it does not make; an office address can be mistaken for a production site. An LLM can turn these partial observations into an apparently coherent assessment. Evaluating the assessment therefore requires an explicit unit of analysis and a record of how evidence and decisions were processed.

This paper studies an implemented LLM evidence-screening workflow. Its observational unit is one frozen source excerpt paired with a requested manufacturing capability and state. The outcome is the model's assessment of page support under a five-component rubric. It is not an externally verified statement about a business. We ask:

- **RQ1:** How are the constructed pairs labeled, and which evidence gaps dominate the nonpositive outputs?
- **RQ2:** How do fieldwise adjudication and literal-quote re-audit change those outputs on the same frozen corpus?

We contribute a documented corpus construction and support rubric, a reconstruction of consequential processing corrections, and an account of what those corrections do and do not validate. The revisions examined here were prompted by failures found during the study. They are retrospective process checks, not a preregistered comparison or a controlled accuracy experiment. Version 0.2 narrows the earlier manuscript's emphasis after the results were known; the original protocols, deviations, and outstanding validation requirements remain part of the record.

2. Related Work

Manufacturing-service discovery already uses knowledge graphs, capability inference, and language models. Li, Liu, and Starly study GNN-based manufacturing capability prediction [1]; Li, Ko, and Ameri integrate graph retrieval with LLM supplier discovery [2]. Qi et al. describe a Web-Knowledge-Web pipeline in which an evolving graph guides subsequent acquisition [3]. The present study evaluates neither a GNN nor a graph-guided discovery policy. It examines the evidence-screening step that could support a later evaluation of such systems.

Search-engine overlap has been studied across thousands of queries [4]. Here it describes the collection boundary, rather than serving as a claim of retrieval quality. Buckley and Voorhees examine evaluation under incomplete relevance judgments [5]; our finite constructed corpus likewise cannot establish relevance outside the observed candidates. Dodge et al. show the importance of documenting web-corpus collection and filtering [6]. Our fetch ledger, excerpt construction, and deviation record expose related choices at a smaller, domain-specific scale.

Zheng et al. evaluate LLM judges against human preferences and document judge biases [7]. Repeated agreement from one model is consequently distinct from external validity. Our contribution is not a new general theory of LLM judging or supplier discovery. It is an empirical audit of how quotation constraints and adjudication logic affected one deployed screening workflow.

3. Corpus and Screening Protocol

3.1 Frozen retrieval and acquisition

On July 30, 2026 (UTC), 40 capabilities from ten manufacturing families were crossed with eight states and three query intents, yielding 960 frozen queries. The intents were direct manufacturer wording, small-batch/job-shop wording, and mixed ownership/locality wording. The last category includes family-owned and local-workshop expressions and is not a firm-size label. The exact lattice, capability definitions, and state list are public.

Each query requested ten web results from Brave and You Search. Brave returned 9,590 ranked rows, with one failed query after the frozen retry policy; You returned 9,600. Transient processing produced 11,715 query-URL union links and 4,638 unique destination URLs. Provider payloads, titles, snippets, and ranks were not retained as the research dataset. The same robots and HTTP workflow was applied to union URLs, and independently fetched publisher-page material was retained in a closed corpus where permitted.

Of the 4,638 URLs, 3,383 returned HTTP 2xx and entered pair construction. There were 1,018 HTTP 403 responses, 129 robots-disallowed flags, and nine hard-timeout flags; process flags can overlap status categories. Access failures therefore affect the observation boundary. Direct acquisition does not imply that a page is first-party or factually corroborated. Retrieval overlap and descriptive subgroup counts appear in Appendix A.

3.2 Observation unit and evidence boundary

The pipeline inherited capability and state from each retrieving query and deduplicated on `(source ID, capability, state)`. Observed intent tags were aggregated. This produced 6,164 pairs from the 3,383 accessible pages. A page may appear in multiple pairs with different requested capabilities or states. Pairs must not be treated as independent companies, and one page need not receive the same label for every task.

Visible text was extracted from the frozen page body. Task keywords and business-context terms selected windows, which were merged and concatenated with omission markers. The extractor budgeted 9,000 source-text characters; separator characters were additional. The reviewer also received the task definition and source-page metadata. It could assign at most one focal business to each pair. Tools and browsing were disabled during review. The target is support in this bounded input, rather than support somewhere on the website or across all pages about the company.

3.3 Five-component support rubric

The retained schema field `eqdp` records whether the excerpt supports all five conditions:

1. A specific operating commercial business is identified.
2. The business directly produces, fabricates, assembles, processes, rebuilds, remanufactures, or performs contract manufacturing.
3. The offering explicitly matches the requested capability.
4. Eligible production presence is supported in the queried state.
5. The offering is current and commercial.

Eligible presence includes an industrial facility or job shop and, where supported, owner/home production. An office, warehouse, service area, registered address, planned plant, or U.S. sales presence with foreign-only production is insufficient. The workflow retains yes, no, and unclear outcomes and component-level decisions. A no or unclear label does not establish that a business lacks the capability; the supplied excerpt may simply fail the evidence test. Literal wording also cannot establish the truth or currentness of the page's claim.

3.4 Repeated review and fieldwise adjudication

A 1,200-pair probability sample was reviewed first. Protocol v0.4 prospectively governed the remaining 4,964 pairs after retrieval and calibration; it was not a pre-retrieval preregistration. The calibration decisions were reused in the full-corpus analysis. Appendix C preserves the sampling and resampling deviations.

Reviewers A and B were separate GPT-5.6-Sol passes with different stance/order prompts. Provider identity, rank, and prior reviewer decisions were hidden from their review prompts. Disagreements across key fields triggered a third same-model pass, C; 2,661 extension pairs required it. Execution failures were retried or requeued rather than assigned unclear labels. These passes were not independent models or human reviewers.

The initial merger selected C's whole record when any key field was disputed. This allowed C to overwrite other fields on which A and B agreed. The corrected rule operates separately for each key field: retain the A/B value when they agree; otherwise use C. This enforces the intended disagreement-only role of adjudication. It does not establish that the retained value is factually correct.

3.5 Literal-quote re-audit and analysis

An audit identified 71 positive rows whose supplied quote was compressed, concatenated, or paraphrased rather than a normalized literal substring of the frozen excerpt. These flagged rows received A/B re-audit from the same evidence with a mandatory contiguous literal quote for a positive decision. Semantic disagreements received fieldwise C adjudication. Unicode, quotation-mark, and whitespace normalization were allowed by the string check. Matching the excerpt verifies a quotation property, not semantic entailment or fidelity to the complete original page.

We report exact label counts for the constructed corpus and reconstruct the processing stages from stored decisions. The fieldwise merge comparison reuses the same A/B/C outputs; the quote stage includes new reviews of the flagged subset. Their differences cannot be interpreted as randomized treatment effects. We report neither factual precision nor recall, and no confidence interval is used to generalize these finite-corpus counts to U.S. manufacturers.

4. Results

4.1 Final labels and repeat-evaluation stability

The final workflow retained **1,148 positive page-support labels**, corresponding to 18.62% of constructed pairs (Table 1). Positive pairs referenced 1,072 destination pages and 796 registrable domains. All retained positive quotes passed the normalized literal-substring check. These are model outputs with an internal quotation check, not verified supplier counts.

Table 1: Final labels on the complete constructed corpus. The denominator is pairs, not businesses.

Final pair label	Count	Share
Page-support yes	1,148	18.62%
No	3,799	61.63%
Unclear	1,217	19.74%
Total	6,164	100%

On the original 4,964-pair extension reviews, before targeted quote re-audit, A/B agreed on `eqdp` for 4,050 pairs (81.59%; nominal Cohen's kappa = 0.609). This measures repeat-evaluation stability under prompt/order variation, not agreement with human experts. The 1,200-pair calibration and extension are kept separate for this stability statistic because they were executed in different stages.

4.2 Evidence gaps in nonpositive outputs

Among the 5,016 nonpositive pairs, insufficient evidence and wrong-state or missing state-production proof accounted for 2,814 primary reasons (56.10%; Table 2). This is a distribution of model-assigned reasons. It does not determine whether the corresponding firms actually lack production capacity or whether supplemental pages could resolve the gap.

Table 2: Primary model-assigned reasons; denominator 5,016. The last row groups the remaining recorded categories. Percentages are rounded.

Primary exclusion reason	Pairs	Share of nonpositive pairs
Insufficient evidence	1,712	34.13%
Wrong state / missing state-production proof	1,102	21.97%
Capability mismatch	568	11.32%
Distributor, reseller, or broker	463	9.23%
Nonmanufacturing service	324	6.46%
Source unavailable within excerpt	290	5.78%
Foreign-only production	203	4.05%
Generic sector claim	133	2.65%
Other recorded reasons	221	4.41%

4.3 Sensitivity to processing rules

Table 3 reconstructs the label trajectory. Whole-record C selection produced 321 unclear labels in fields where A and B had agreed on no: 38 in calibration and 283 in the extension. Fieldwise merging restored those no labels and left the positive count unchanged. These are changes to the primary `eqdp` field. Across all ten adjudicated key fields, 339 rows changed, including 18 with no change to the primary label.

Table 3: Retrospectively reconstructed processing stages, each totaling 6,164 pairs. The first transition changes merge logic on stored reviews; the second includes re-review of 71 flagged positives. This is not an accuracy comparison.

Processing stage	Yes	No	Unclear
Whole-record C merge, before quote re-audit	1,179	3,478	1,507
Fieldwise merge, before quote re-audit	1,179	3,799	1,186
Fieldwise merge and literal-quote re-audit	1,148	3,799	1,217

Of those 71 flagged positives, 40 remained yes with acceptable literal quotations and 31 became unclear. The positive count fell from 1,179 to 1,148. It would be incorrect to describe all 71 as false positives or to treat the 31 downgraded cases as independently established errors. The evidence supports a narrower result: enforcing the revised quote requirement, together with re-review, changed both the retained quotations and some labels.

A separate size-screen failure reinforced the need to distinguish source material from earlier model output: an initial pass included prior model rationales in its evidence context. That pass was discarded and regenerated from frozen page text. Because the input procedure changed, we do not interpret differences from the invalid run as measured accuracy gains. Its corrected, exploratory results remain in Appendix B.

5. Interpretation and Limitations

Agreement, quotation integrity, and validity are different properties. Fieldwise merging prevents an adjudicator from changing agreed fields under the stated rule. Literal matching checks whether the reported words occur in the supplied excerpt. Neither property establishes that the words entail all five support conditions, that the webpage is truthful, or that the model agrees with a domain expert. The corrections make specific processing failures inspectable without substituting internal checks for external validation.

The task definition limits downstream use. A page can fail this single-excerpt test while another page supplies the missing location evidence. A future system that combines pages would require a multi-page reference task and identical evidence access for its baselines. Likewise, a graph-based candidate-ranking experiment would need an independent human test set and controls for extra text, simple graph statistics, and message passing. This corpus is not already a validated GNN benchmark.

Collection and model scope are narrow. The study uses two providers, selected capability/state templates, accessible top-10 destination pages, and one model with a particular excerpt procedure. Repeated pages and entities induce dependence. Fetch failures, truncation, query wording, stale pages, and one-focal-business extraction shape the observed labels. Results do not estimate national coverage, industry prevalence, procurement readiness, actual buyer-supplier links, or SME status. No comparative accuracy claim against another model or reviewer is supported.

The audit is retrospective and incompletely validated. Processing failures motivated the corrections, and this manuscript refocus was made after results were known. The changes were not prespecified experimental arms. The original human double-annotation and entity-resolution precision gates remain unmet. Removing entity-level findings from the main narrative does not satisfy or waive those gates. The recorded calibration sampling mismatch remains a limitation; its exploratory interval is not a main result here.

Public artifacts expose only part of the evidence chain. The public repository contains code, schemas, queries, logs, and aggregate bytes, while the frozen page bodies and row-level decisions remain closed. Hashes establish byte identity, not factual correctness. Independent researchers can inspect the workflow and aggregates but cannot reproduce or audit individual frozen judgments from the public package alone.

6. Conclusion

On 6,164 constructed source-capability-state pairs, the final same-model workflow assigned 1,148 positive page-support labels. Retrospective reconstruction showed that whole-record adjudication and nonliteral quotations materially affected its outputs: fieldwise merging restored 321 agreed no labels, and re-audit downgraded 31 flagged positives. These findings identify concrete process sensitivities in LLM-assisted evidence screening. Human validation is required before interpreting the labels as benchmark judgments or measuring factual screening performance.

Data, Code, and AI-Use Disclosure

Frozen queries, processing and review scripts, schemas, deviation logs, and the row-free aggregate package are publicly archived in the US-SME Signal Graph repository. The original aggregate package is anchored at commit c036829; its eight published SHA-256 entries were checked after upload. The package remains versioned as *long-tail-evidence-audit-v0.1-exploratory* because this manuscript revision does not change its data. Search-provider payloads and fetched third-party page text are excluded from the public artifacts. The companion processing-stage summary records the reconstructed counts and input fingerprints. Its reconstruction script checks all 6,164 final rows across ten key fields against the stored final table and recomputes the 71 quote-check failures. Re-running that audit requires the closed corpus; the public summary does not enable record-level reproduction.

GPT-5.6-Sol generated the study's screening decisions through the disclosed repeated-pass workflow. An AI coding assistant assisted with analysis checks, manuscript revision, and formatting. These activities do not constitute independent human review. Author review and the selected venue's AI-use requirements remain outstanding submission tasks. The revision record and claim audit preserve the research scope change and validation status.

References

[1] Y. Li, X. Liu, and B. Starly. Manufacturing service capability prediction with Graph Neural Networks. *Journal of Manufacturing Systems*, 74, 291-301, 2024. https://doi.org/10.1016/j.jmsy.2024.03.010.

[2] Y. Li, H. Ko, and F. Ameri. Integrating Graph Retrieval-Augmented Generation With Large Language Models for Supplier Discovery. *Journal of Computing and Information Science in Engineering*, 25(2), 021010, 2025. https://doi.org/10.1115/1.4067389.

[3] Y. Qi, Y. Qi, and T. Wagh. Coverage-Aware Web Crawling for Domain-Specific Supplier Discovery via a Web-Knowledge-Web Pipeline. *arXiv:2602.24262v4*, 2026. https://arxiv.org/abs/2602.24262v4.

[4] N. Yagci, S. Sünkler, H. Häußler, and D. Lewandowski. A Comparison of Source Distribution and Result Overlap in Web Search Engines. *Proceedings of the Association for Information Science and Technology*, 59(1), 346-357, 2022. https://doi.org/10.1002/pra2.758.

[5] C. Buckley and E. M. Voorhees. Retrieval Evaluation with Incomplete Information. *SIGIR '04*, 25-32, 2004. https://doi.org/10.1145/1008992.1009000.

[6] J. Dodge, M. Sap, A. Marasović, W. Agnew, G. Ilharco, D. Groeneveld, M. Mitchell, and M. Gardner. Documenting Large Webtext Corpora: A Case Study on the Colossal Clean Crawled Corpus. *EMNLP*, 1286-1305, 2021. https://aclanthology.org/2021.emnlp-main.98/.

[7] L. Zheng et al. Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena. *Advances in Neural Information Processing Systems*, 36, 46595-46623, 2023. https://papers.nips.cc/paper_files/paper/2023/hash/91f18a1287b398d378ef22505bf41832-Abstract-Datasets_and_Benchmarks.html.

Appendix A. Collection Context and Descriptive Subgroups

The states were Massachusetts, Pennsylvania, Michigan, Ohio, North Carolina, Texas, Arizona, and California. Ten capability families supplied four capabilities each. Cross-query deduplication and pair construction mean that ranked rows, URLs, pairs, and businesses are different units. Query-URL overlap was 63.81% when pooled across all queries and 65.55% as the macro mean of per-query Jaccards. On the 959 dual-success queries, the corresponding values were 63.86% and 65.62%. Overlap measures returned-URL agreement, not producer recall or provider quality.

Table 4: Descriptive label fractions in selected query families. They are not national industry estimates or controlled industry effects.

Industry family	Yes	Pairs	Positive-label rate
Precision metal	294	645	45.58%
Wood / industrial packaging	223	601	37.10%
Remanufacturing	139	533	26.08%
Electronics	113	551	20.51%
Additive manufacturing / tooling	126	715	17.62%
Polymers / composites	110	622	17.68%
Contract consumer production	61	582	10.48%
Industrial textiles	50	605	8.26%
Semiconductor	15	590	2.54%
Battery	17	720	2.36%

State-conditioned fractions ranged from 11.46% in Massachusetts to 24.68% in Texas. Intent tags overlap: direct-observed pairs had 674 positives among 3,164 incidences, mixed ownership/locality 695 among 3,009, and small-batch 491 among 2,745. These are descriptive incidences, not randomized wording effects. Full state and intent tables remain in the public aggregate package.

Appendix B. Provisional Entities and Destination-Page Size Screen

These analyses are retained as exploratory context and are not primary contributions of this revision. Positive pairs were initially blocked by domain, normalized name, and state. Dedicated business domains could support initial grouping; shared third-party directories could not. Following quote correction, 89 flagged domain clusters and 11 fuzzy/alias candidate pairs received same-model review, alongside exact normalized-name/state consolidation across domains. The 1,148 positive pairs mapped to 790 provisional operating-entity clusters and 785 corporate-group clusters. False-merge and false-split uncertainty has not been measured externally, and clusters were not equated with production sites.

The corrected size screen used only frozen destination-page text around employee, workforce, owner-only, nonemployer, and federal-program terms. It replaced an invalid pass containing earlier model rationales. Empty contexts were screen-negative: 446 positive-pair contexts contained a size term and 702 did not. Thirty-nine surviving pair-level signals mapped to 29 provisional clusters for A/B review and fieldwise C adjudication on five semantic disagreements. All 29 eligible entity-level quotations passed literal matching; 761 automatic screen negatives were not quote-audit eligible.

Table 5: Nested or overlapping categories, not an additive partition. The 23 numerical cases comprise 14 single-band and nine cross-band cases. All counts depend on provisional entity resolution.

Size-screen result	Provisional clusters
Nonhistorical numerical employee evidence	23
Wholly within one descriptive employee band	14
Band 10-99	10
Band 100-499	2
Band 500+	2
Numerical bounds crossing bands	9
Current owner-only support	0
Explicit SAM/SBA or named-program page representation	3
No nonhistorical numerical employee evidence	767

The three federal representations comprise two explicit SAM/SBA claims and one named-program claim; one overlaps a single-band case. They are page representations, not independent determinations. The descriptive employee bands are not SBA size definitions. External SBA-small verification was not performed. This narrow screen measures evidence on already retrieved destination pages under the term lexicon; it cannot estimate firm-size information availability across the public web.

Appendix C. Deviations and Outstanding Validation

Calibration. The 1,200-pair calibration ended with 214 yes, 677 no, and 309 unclear labels. Its sampler used 30 industry-family by primary-intent strata, narrower than protocol v0.3's stated balancing dimensions. Its 10,000-replicate bootstrap resampled pairs within strata instead of query units. The reported weighted estimate, 17.84%, and interval, 15.84%-19.86%, remain historical exploratory diagnostics in the aggregate package; they are not presented as protocol-conformant or used as confirmatory evidence here.

Audit trail. Protocol v0.4 was committed after retrieval and calibration, before the 4,964-pair extension. The public model-review deviation log records the human-review substitution, amendment timing, whole-record merge, quote-denominator correction, overlap terminology, literal-quote re-audit, and sampling mismatch. The v0.2 revision record documents this retrospective editorial refocus. None of these records converts the original study into a fully preregistered or human-validated experiment.

Submission Gates

The existing study requirements remain in force. This draft is not publication-ready until the outstanding gates in the repository's claim audit are completed:

1. At least 300 provider-blind cases receive independent human double annotation, component and overall agreement are reported, and the preregistered $\kappa \geq 0.70$ gate is evaluated. Model-versus-human comparison must include no and unclear cases, not only model positives.
2. The presampled 200-pair entity-resolution precision audit is executed and its ≥ 0.95 gate is evaluated. Moving entity results to an appendix does not complete or waive this requirement.
3. Corrected entity-size tables and hashes remain synchronized; the selected venue's literature coverage, bibliography, and AI-use requirements are checked.
4. Any future confirmatory use of the calibration interval reconciles sampling and resampling with the declared protocol. This revision makes no such claim.
5. The public aggregate package and post-publication checksum verification remain documented. This archival step has been completed; it provides no substitute for human validation.

No new human labels, trained GNN, external entity verification, or submission outcome is reported in this revision.